Parametric Equations

Questions 1–66

2.41 — Question 1. Prerequisite knowledge: Those questions that involve trigonometry use radians
For each of the following, complete the table, plot the coordinates and sketch the curve for $-2 \leq t \leq 2$.

t	-2	-1	0	1	2
x					
y					

a) $x = t^2$ and $y = 2t$
b) $x = t + 2$ and $y = -2t^2 + 1$
c) $x = t(t + 2)$ and $y = t^2(t - 1)$
d) $x = t(1 - t)$ and $y = 2t^2$
e) $x = \frac{4}{2+t^2}$ and $y = t(t + 1)$

2.41 — Question 2. Given that $x = t$, find y in terms of t for each of the following pairs of parametric equations:
a) $y = 2x^2$
b) $y = x^2 + 4$
c) $y = e^x + 3$
d) $y - 2x = 8$
e) $\frac{x^2}{y} = 1$
f) $x^3y = -10$
g) $x^2 + y^2 = 1$

2.41 — Question 3. Find the cartesian equation (with y as the subject) for each of the following curves:

a) $x = 2t$ and $y = t^2$
b) $x = t - 1$ and $y = 2t + 3$
c) $x = t - 3$ and $y = e^t - t$
d) $x = -2t$ and $y = t^2 \ln t$

2.41 — Question 4. Find the cartesian equation (with x as the subject) for each of the following curves:

a) $x = t^2$ and $y = t - 1$
b) $x = t^3 - t$ and $y = \frac{t}{2}$
c) $x = t^5$ and $y = 3t$
d) $x = e^{t-1}$ and $y = t + 5$

2.41 — Question 5. For each of the following curves, find the coordinates of intersections with the axes:

a) $x = t + 1$ and $y = t - 3$
b) $x = \cos t$ and $y = 2\sin t$ for $0 \leq t < 2\pi$
c) $x = 2\cos t$ and $y = \sin 3t$ for $0 \leq t < 2\pi$
d) $x = t + 5$ and $y = \ln(t + 1)$
e) $x = t^3 - 3t^2 - 6t + 8$ and $y = e^{-2t}$

2.41 — Question 6. A curve is given by parametric equations

$$x = t^2 \quad \text{and} \quad y = t^3$$

Given the curve may be written in the form $y^2 = x^k$, state the value of k.

2.41 — Question 7. Find the cartesian equation for each of:

a) $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$

b) $x = t - \frac{2}{t}$ and $y = t + \frac{2}{t}$

c) $x = 7t + \frac{4}{t}$ and $y = 7t - \frac{4}{t}$

2.41 — Question 8. A curve is given by parametric equations

$$x = \frac{1}{t} - 1 \quad \text{and} \quad y = \frac{2+3t}{1+3t}$$

Show that the cartesian equation of the curve can be written in the form $y = \frac{ax+b}{x+4}$, $x \neq -4$ where a and b are integers to be found.

2.41 — Question 9. Find a pair of parametric equations for each of the following cartesian equations:

a) $4x^2 + 9y^2 = 1$
b) $3x^2 + 2y^2 = 1$
c) $5x^2 + \frac{1}{2}y^2 = 1$
d) $4x^2 + \frac{2}{3}y^2 = 2$

2.41 — Question 10. Show that each of the following may be written in the form

$$(x - p)^2 + (y - q)^2 = k$$

where p, q and k are integers and $0 \leq \theta < 2\pi$:

a) $x = 3\cos\theta$ and $y = 3\sin\theta$
b) $x = 5\sin\theta$ and $y = -5\cos\theta$
c) $x = \cos\theta + 1$ and $y = \sin\theta - 3$
d) $x = \sin\theta - 4$ and $y = \cos\theta + 8$

2.41 — Question 11. A fireworks display is planned so that a sequence of firework bursts follow a trajectory in the sky, in metres, such that

$$x = 10\cos t \quad \text{and} \quad y = 15 + 5\sin t$$

for $0 < t \leq 2\pi$ where t represents the position of a burst of a firework at time t seconds. Find

a) the cartesian equation of the trajectory traced by the fireworks.
b) the maximum height reached by the fireworks in one cycle of motion.
c) the value of t which provides the maximum height in part b).
d) the period of the fireworks sequence.

2.41 — Question 12. A curve is defined for $0 \leq \theta < 2\pi$ by

$$x = 2k + 2k\cos\theta \quad \text{and} \quad y = 8 - 3\sin\theta$$

for $k > 0$.

a) Find a cartesian equation for the curve.

It is given that the curve is a circle. Find

b) the value of k.
c) the centre of the circle.
d) the radius of the circle.

2.41 — Question 13. Prerequisite knowledge for Q13: Trig Identities Write the following

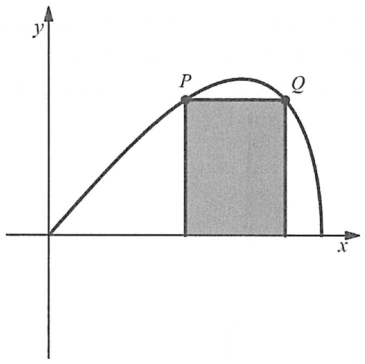
$$x = 2\tan\theta \text{ and } y = 2 + \sec\theta, \quad 0 \leq \theta < 2\pi$$

as a cartesian equation.

2.41 — Question 14. A curve is defined by the parametric equations

$$x = \sqrt{3}\sin\theta, \quad y = \sin 2\theta$$

for $0 \leq \theta \leq \frac{\pi}{2}$.



The points P and Q lie on the curve. The line $x = \frac{3}{2}$ meets the curve at the point Q.

a) Find the coordinates of P and Q.

b) Hence find the exact area of the rectangle bounded by the x-axis and the points P and Q.

2.41 — Question 15. Find the cartesian form of the curve which satisfies all the points on:

a) $x = \cos\theta$ and $y = \sin^2\theta$ for $0 \leq \theta \leq \pi$

b) $x = \sin\theta$ and $y = \tan\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

2.41 — Question 16. A curve is defined by

$$x = \frac{t^2+7}{t^2+1} \quad \text{and} \quad y = \frac{6t}{t^2+1}$$

a) Show that the curve defined by $t \in \mathbb{R}$ is a circle with centre $(4, 0)$.

b) Find the radius of the circle.

2.41 — Question 17. For each of the following parametric curves, find all the points of intersection with the circle given by

$$x^2 + y^2 = 100$$

where $0 \leq \theta \leq 2\pi$. Give your answers in exact form.

a) $x = 10\cos\theta$, $y = 4\sin\theta$
b) $x = 2\cos\theta$, $y = 10\sin\theta$
c) $x = 6\sqrt{2}\cos\theta$, $y = 2\sqrt{46}\sin\theta$

2.41 — Question 18. Prerequisite knowledge for Q18: Double Angle Formulae A curve is given by parametric equations

$$x = 5 + 2\cos\theta \quad \text{and} \quad y = -1 + 2\cos 2\theta$$

for $0 \leq \theta \leq \pi$.

a) Find the cartesian equation of the curve which satisfies all the points on the curve.
b) State the domain and range of the curve.

The line $y + x = k$ where k is a constant, intersects the curve at two distinct points.

c) State the range of the values of k giving your answer in interval notation.

2.41 — Question 19. Prerequisite knowledge for Q19–Q21: Domain and Range of Functions Find the largest possible domain and range of:

a) $x = t^2 + 1$ and $y = \sqrt{t}$
b) $x = 2t^2$ and $y = 4 - \sqrt{t}$
c) $x = \sin t$ and $y = \cos t$
d) $x = \sin 3t$ and $y = \cos(t + \frac{\pi}{2})$
e) $x = e^t$ and $y = e^{-2t}$
f) $x = \ln t$ and $y = t^2$

2.41 — Question 20. By first considering valid values for t, find the largest possible domain and range of the curve defined by $x = \sin\sqrt{t}$ and $y = \frac{1}{t}$.

2.41 — Question 21. For $0 \leq \theta < 2\pi$ a curve is defined by

$$x = \sin 2\theta \quad \text{and} \quad y = 4\cos(\theta - \tfrac{\pi}{6})$$

a) Find the domain and range of the curve.
b) Find the coordinates where the curve meets the coordinate axes.
c) Hence, sketch the curve.

2.42 — Question 1. By finding the cartesian equation of the curve, given by

$$x = t - 1 \qquad \text{and} \qquad y = 2t^2$$

find $\frac{dy}{dx}$ in terms of x.

2.42 — Question 2. Find $\frac{dy}{dx}$ in terms of t for each of the following:

a) $x = t + 1$ and $y = -2t^2 + 2$

b) $x = t(t + 3)$ and $y = t^2(t + 1)$

c) $x = e^{2t}$ and $y = 2e^t$

d) $x = 4e^{2t}$ and $y = 3e^{-t} + 1$

e) $x = t - \cos t$ and $y = t - \sin t$

2.42 — Question 3. Prerequisite knowledge for Q3: Quotient Rule Find $\frac{dy}{dx}$ in terms of t for $x = \frac{1}{1+t^2}$ and $y = t^2$.

2.42 — Question 4. It is given that $x = t + 1$, $y = t^2$, where $t \geq 0$. Find the equation of the tangent to the curve when $t = 2$.

2.42 — Question 5. It is given that $x = t^3$, $y = 2t$, where $t \geq 0$. Find the equation of the normal to the curve when $t = 3$.

2.42 — Question 6. It is given that $x = e^t$, $y = \frac{1}{t}$, where $t > 0$. Find the equation of the normal to the curve when $t = 1$.

2.42 — Question 7. For each of the following curves defined parametrically, find the coordinates of the stationary point(s) of the curve:

a) $x = t^4 + t$ and $y = 10t - 5t^2$ where $t \geq 0$

b) $x = 6t$ and $y = t^3 - 5t^2 + 3t$ where $t \geq 0$

c) $x = e^{2t}$ and $y = 6t^2 - \frac{4}{3}t^3 - 5t$ where $t \geq 0$

2.42 — Question 8. A curve is defined by the parametric equations

$$x = 3 - t^3 \qquad \text{and} \qquad y = (t+2)^2$$

for $-1 \leq t \leq 1$. Find

a) the coordinate of the point with the maximum y-value.
b) the coordinate of the point with the minimum y-value.
c) $\frac{dy}{dx}$ in terms of t.

2.42 — Question 9. A curve is defined by the parametric equations

$$x = \frac{4}{t} \qquad \text{and} \qquad y = 3t - 1$$

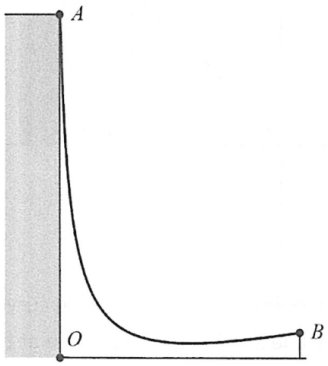
a) Find the equation of the tangent line to the curve at the point $(2, 5)$ giving your answer in the form $y = mx + c$.
b) Hence find the area bounded by the tangent line at $(2, 5)$ and the coordinate axes.

2.42 — Question 10. Given that $t > 0$, find the equation of the normal to the curve given by

$$x = 1 - \frac{1}{t} \quad \text{and} \quad y = 2t(t-1)$$

when $x = -2$, giving your answer in the form $ax + by = c$ where a, b and c are integers to be found.

2.42 — Question 11. A drop slide is shown in the diagram. Point A is at the start of the slide and point is B at the end of the slide. Point O lies on the ground.



It may be modelled by parametric equations

$$x = \tfrac{1}{2}t^2 - \tfrac{1}{8} \quad \text{and} \quad y = \frac{5}{t} + t - \frac{21}{5}$$

for $0.5 \leq t \leq 3$. The horizontal distance from O is x metres and the vertical distance from O is y metres. According to the model, find

a) the height at the start of the slide.
b) the height at the end of the slide.
c) how far horizontally point B is from the start of the slide.
d) the difference in height between the start of the slide and the lowest point of the slide, giving your answer to the nearest centimetre.

2.42 — Question 12. A curve is defined by the parametric equations

$$x = t + a \qquad \text{and} \qquad y = t^2 + 3$$

where $-3 \leq t \leq 3$ and a is an integer. Find

a) the cartesian curve which satisfies all the points on the curve.

b) the equation of the normal to the curve when $t = 1$ in the form $y = mx + c$, giving your answer in terms of a.

2.42 — Question 13. A curve is defined by the parametric equations

$$x = \sqrt{3}\sin\theta \qquad \text{and} \qquad y = 2\sin 2\theta$$

for $0 \leq \theta \leq \frac{\pi}{2}$.

a) Find $\frac{dy}{dx}$ in terms of θ.

b) Show that the equation for the normal line to the curve at $\theta = \frac{\pi}{3}$ may be written in the form

$$a\sqrt{3}y - bx = c$$

where a, b and c are integers to be found.

c) Find the area bound by the coordinate axes and the normal line to the curve when $\theta = \frac{\pi}{3}$.

2.42 — Question 14. A curve is defined by the parametric equations

$$x = 2 \times 3^{-t} + 4 \qquad \text{and} \qquad y = 4 \times 3^{t} - 6$$

for $t \in \mathbb{R}$.

a) Show that $\frac{dy}{dx} = -2 \times 3^{2t}$

b) Show that the curve does not cross the y-axis.

c) Show that the cartesian equation which satisfies all the points on the curve may be written in the form

$$y = \frac{a}{x - b} - c$$

where a, b and c are integers and $x > b$.

d) As $t \to \infty$, $x \to \alpha$. Find the value of α.

e) As $t \to -\infty$, $y \to \beta$. Find the value of β.

f) Find where the curve crosses the x-axis.

g) Hence sketch the curve.

2.42 — Question 15. Prerequisite knowledge for Q15: Reciprocal Trigonometric Functions and Chain Rule A curve is defined by

$$x = \sin 4\theta \qquad \text{and} \qquad y = \sec^2 \theta$$

for $0 \leq \theta \leq \frac{\pi}{2}$.

a) Find $\frac{dy}{dx}$ in terms of θ.

b) Hence find the exact value of the gradient of the tangent of the curve when $y = 2$.

2.42 — Question 16. A curve is defined by parametric equations

$$x = 5t^3 \quad \text{and} \quad y = t^4$$

for $t \geq 0$.

a) Show that the equation of the tangent at the point t on the curve is given by

$$15y - 4tx + 5t^4 = 0$$

A triangle is bound by the x-axis, the tangent line to the curve at $x = 40$ and the line $x = 40$.

b) Find the area of the triangle.

2.42 — Question 17. A curve is defined by parametric equations

$$x = \tan\theta + 1 \qquad \text{and} \qquad y = \sec\theta$$

for $-\frac{\pi}{3} \leq \theta \leq \frac{\pi}{3}$. The point P lies on the curve when $\theta = a$ where $-\frac{\pi}{3} \leq a \leq \frac{\pi}{3}$.

a) Find the equation for the normal to the curve at the point P in the form $y = mx + c$ giving your answer in terms of a.

b) Hence find the minimum point on the curve.

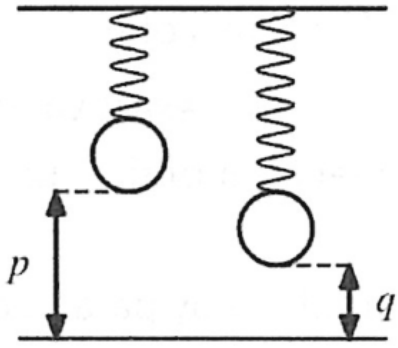
2.42 — Question 18. A curve is defined by

$$x = t^4 \quad \text{and} \quad y = 2t^3(1+t)$$

where $t \in \mathbb{R}$.

a) Find $\frac{dy}{dx}$ in terms of t.
b) Find the coordinates of the minimum point of the curve.
c) Find where the curve crosses the y-axis.
d) Find where the curve crosses the x-axis.
e) Hence sketch the curve.

2.42 — Question 19. Prerequisite knowledge for Q19–Q21: Harmonic forms Two springs are attached to a ceiling with masses attached. They oscillate up and down, between two heights above the floor, p and q metres as shown in the diagram.



They may be modelled using the parametric equations

$$p = \tfrac{1}{2} - \tfrac{1}{2}\cos t \quad \text{and} \quad q = \tfrac{1}{2} - \tfrac{1}{4}\sin t$$

where $0 \leq t \leq 8\pi$.

a) Find the initial height difference between the two springs.
b) Write the difference in height between them in the form $R\sin(t-\alpha)$ where R and α are positive constants to be found and $0 < \alpha < \frac{\pi}{2}$.
c) Find the maximum distance between the two springs.

2.42 — Question 20. A curve is defined by

$$x = 4\cos\theta \quad \text{and} \quad y = 5\sin\theta$$

for $0 \leq \theta \leq 2\pi$. The tangent lines to the curve at two points, A and B, pass through the point $(4, 12)$.

a) Find $\frac{dy}{dx}$ in terms of θ.

b) State the equation of the tangent line in terms of θ.

c) Hence show that values of θ at A and B satisfy the equation

$$12\sin\theta + 5\cos\theta = 5$$

d) Hence find the two possible values of θ.

2.42 — Question 21. A curve is defined by

$$x = a\cos\theta \quad \text{and} \quad y = 4a\sin\theta$$

a) Show that the equation of the tangent line is

$$y\sin\theta + 4x\cos\theta = 4a$$

The tangent line passes through the point $(a, 3a)$.

b) For $0 < \alpha < \frac{\pi}{2}$, show that θ satisfies

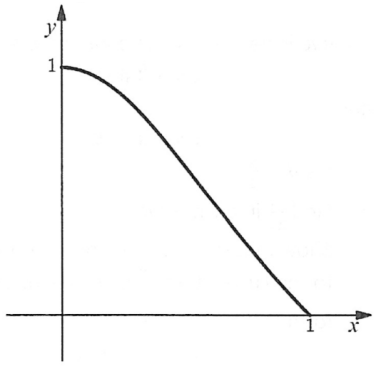
$$5\sin(\theta + \alpha) = 4$$

c) Hence find the two possible values of θ when $0 \leq \theta < 2\pi$.

2.42 — Question 22. Prerequisite knowledge for Q22: The Newton-Raphson Method and Double Angle Formulae The curve C shown below is defined by parametric equations

$$x = \tan t \quad \text{and} \quad y = \cos 2t$$

for $0 \leq t \leq \frac{\pi}{4}$.



a) Find the equation of the line that connects the points on the curve C when $t = 0$ and $t = \frac{\pi}{4}$.
b) Show that $\frac{dy}{dx} = -4 \sin t \cos^3 t$
c) Use Newton-Raphson's method starting at $x_0 = 0$ to show that a solution to

$$4 \sin x \cos^3 x = 1$$

occurs when $x = 0.2874$ (to 4 significant figures).
d) Hence, find the range of values of k such that the line $y + x = k$ intersects the curve at least once.

2.42 — Question 23. Prerequisite knowledge for Q23: Product Rule and Small Angle Approximations A curve is defined by the parametric equations

$$x = \tfrac{1}{8} \sin t \quad \text{and} \quad y = t \cos t$$

where $0 \leq t \leq \pi$.
a) Find $\frac{dy}{dx}$ in terms of t.
b) Given that the stationary point is close to the origin, find the stationary point of the curve by using small angle approximations for $\cos t$ and $\sin t$.

2.43 — Question 1. Prerequisite knowledge: The following questions may involve integrating standard functions or reversing the chain rule. For each of the following curves defined by the parametric equations, find $\int y\,dx$:

a) $x = 3t^3 - t$ and $y = t^2 + 1$
b) $x = e^t$ and $y = e^{-t} + 1$
c) $x = 3^t$ and $y = 9^t$

2.43 — Question 2. For each of the following curves, where the lower and upper limit (t_1 and t_2 respectively) of the definite integral have already been found in terms of t, find $\int_a^b y\,dx$ giving your answers in exact form:

a) $x = 2t^3 + t$ and $y = t^2 + 4$ between $t_1 = 1$ and $t_2 = 2$
b) $x = t^2$ and $y = e^t$ between $t_1 = 0$ and $t_2 = 2$
c) $x = 2^t$ and $y = 4^t$ between $t_1 = 0$ and $t_2 = 1$
d) $x = -\ln t$ and $y = t^3$ between $t_1 = 0.5$ and $t_2 = 1$

2.43 — Question 3. A curve is defined by parametric equations

$$x = t^2 \quad \text{and} \quad y = \frac{1}{t}$$

for $1 \leq t \leq 2$.

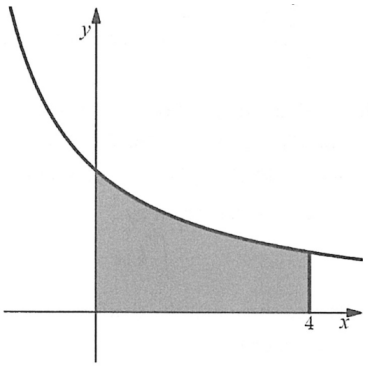
a) Find the coordinates of the curve when $t = 1$ and $t = 2$.

b) Sketch the curve.

c) Find the area enclosed by the curve, the coordinate axes, the lines $x = 1$ and $x = 4$.

2.43 — Question 4. The diagram shows part of a curve defined by

$$x = 2t - 3 \quad \text{and} \quad y = \frac{4}{t}$$



The shaded region is bounded by the curve, the coordinate axes and the line $x = 4$.

a) Find the value of t when $x = 0$ and $x = 4$.

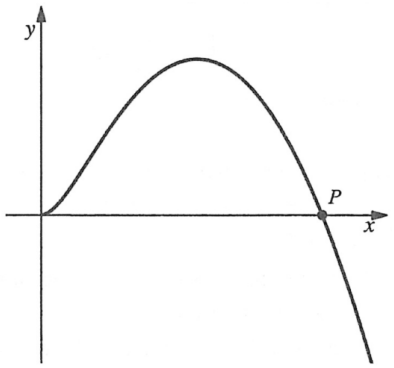
Find the area of the shaded region

b) by using parametric integration.

c) by first finding a cartesian equation for the curve.

2.43 — Question 5. A curve is defined by parametric equations

$$x = t^2 + 2t \quad \text{and} \quad y = 2t^2 - t^3 \quad \text{for } t \geq 0$$



The point P is where the curve meets the x-axis at $x = k$. The area A is enclosed by the curve and the x-axis is given by

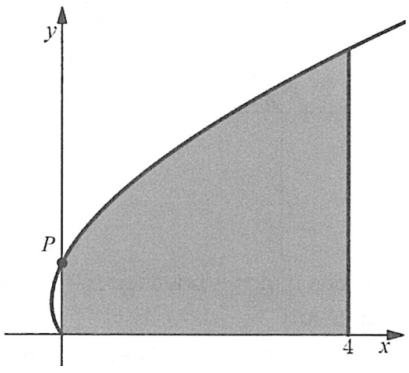
$$A = \int_0^k y \, dx$$

a) Find the value of k.
b) Find the area A.

2.43 — Question 6. The diagram shows the curve given by

$$x = t^2(t - 1) \quad \text{and} \quad y = t^2 \quad \text{for } t \geq 0$$

The point P lies on the y-axis. The curve meets the y-axis when $t = 0$ and at point P.



a) Find the value of t at P.
b) Find the area of the shaded region, which is bounded by the curve, the positive x-axis, the lines $x = 0$ and $x = 4$.

2.43 — Question 7. A curve is defined by parametric equations

$$x = \frac{t}{t+4} \qquad \text{and} \qquad y = \frac{8}{(t+4)^2}$$

where $0 \leq t \leq 5$.

a) Find the coordinates of the curve when $t = 0$ and when $t = 5$.

b) Show that the area enclosed by the curve, the axes and the line $x = \frac{5}{9}$ is given by

$$\int_0^5 \frac{32}{(t+4)^4}\, dt$$

c) Hence, find the exact area.

2.43 — Question 8. The parametric equations of a curve are

$$x = \tan\theta \quad \text{and} \quad y = \cos^3\theta$$

where $0 \leq \theta < \frac{\pi}{2}$.

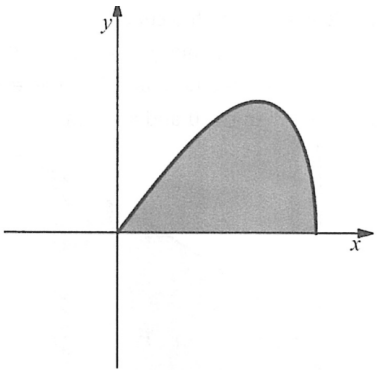
a) Find the coordinates of the curve when $x = 0$ and when $x = \sqrt{3}$.

b) Show that the area enclosed by the curve, the x-axis and the line $x = \sqrt{3}$ is $\frac{\sqrt{3}}{2}$.

2.43 — Question 9. The diagram shows part of the curve defined by parametric equations

$$x = 3\sin t \quad \text{and} \quad y = 2\sin 2t$$

for $0 \leq t \leq \frac{\pi}{2}$.



The shaded region is bounded by the curve and the x-axis.

a) Show that the shaded area is given by

$$\int_0^{\frac{\pi}{2}} 12\sin t\cos^2 t\, dt$$

b) Hence show that the shaded area has an exact area of 4.

2.43 — Question 10. A curve is defined by the parametric equations

$$x = t^3 \quad \text{and} \quad y = k\sqrt{t}$$

where $0 \leq t \leq 1$ and k is a positive constant. It is known that the area between the curve, the x-axis and the lines $x = 0$ and $x = 1$ is 3.

a) Show that $k\int_0^1 3t^{\frac{5}{2}}\, dt = 3$.

b) Find the value of k.

2.43 — Question 11. A curve is defined by parametric equations

$$x = t + \sin t \qquad \text{and} \qquad y = \tfrac{1}{2}\cos^2 t$$

where $0 \leq t \leq \pi$.

a) Find the coordinates of the curve when $t = 0$ and when $t = \pi$.

b) Show that the area enclosed by the curve, the axes and the line $x = \pi$ is given by

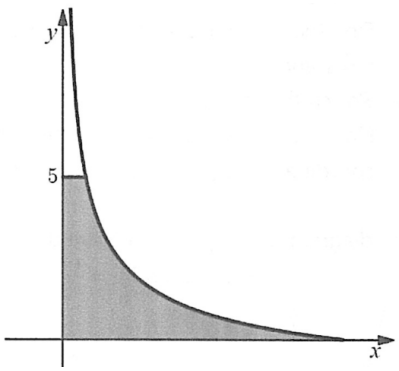
$$\frac{1}{2}\int_0^{\pi} (1 - \sin^2 t + \cos t - \cos t \sin^2 t)\, dt$$

c) Hence, find the exact area.

2.43 — Question 12. The diagram shows part of the curve defined by parametric equations

$$x = t^2 \quad \text{and} \quad y = \frac{6}{t} - 2$$

for $0 \leq t \leq 3$.



The shaded region is bounded by the curve, the coordinate axes and the line $y = 5$.

a) By considering $\int_0^5 x\, dy$, show that the shaded area is given by

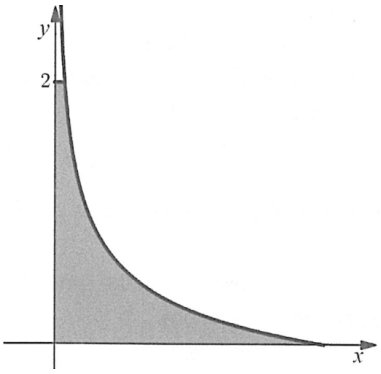
$$\int_{\frac{6}{7}}^{3} 6\, dt$$

b) Hence show that the shaded area has an exact area of $\frac{90}{7}$.

2.43 — Question 13. The diagram shows part of the curve defined by parametric equations

$$x = 5t^3 \quad \text{and} \quad y = \frac{1}{t} - 1$$

for $0 \leq t \leq 1$.



The shaded region is bounded by the curve, the coordinate axes and the line $y = 2$.

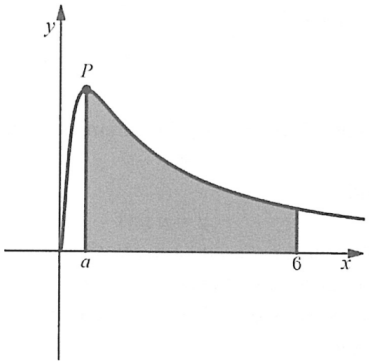
a) Show that the shaded area is given by $-\int_{\frac{1}{3}}^{1} 5t\,dt$.

b) Hence show that the shaded area has an exact area of $\frac{20}{9}$.

2.43 — Question 14. Prerequisite knowledge for Q14–Q22: Double Angle Formulae, Integration by Parts, Product Rule, Partial Fractions, Integration by Substitution The diagram shows the curve given by

$$x = \frac{6}{t^2} \quad \text{and} \quad y = t^3 e^{-t}$$

where $t > 0$. The maximum of the curve is at the point P with $x = a$ where a is a constant to be found.



a) Find the coordinates of P.

b) Hence, show that the area enclosed by the curve, the x-axis and the line $x = a$ and $x = 6$ is

$$\frac{12}{e}\left(1 - \frac{1}{e^2}\right)$$

2.43 — Question 15. Find the area under the curve defined parametrically by

$$y = \cos\theta, \quad x = \sin\theta$$

between $\theta = 0$ and $\theta = \frac{\pi}{2}$.

2.43 — Question 16. Given

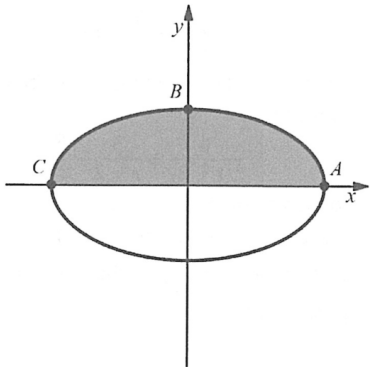
$$x = 2\sin t \quad \text{and} \quad y = 4\cos t$$

find $\int_0^2 y\,dx$ giving your answer in exact form.

2.43 — Question 17. The diagram shows the curve defined by parametric equations

$$x = 9\cos\theta \quad \text{and} \quad y = 5\sin\theta$$

for $0 \leq \theta < 2\pi$.



The curve meets the x-axis at points A and C. The curve meets the positive y-axis at the point B.

a) Find the coordinates of A, B and C and hence the value of θ at each of these points.

b) Show that the shaded region enclosed by the ellipse is given by

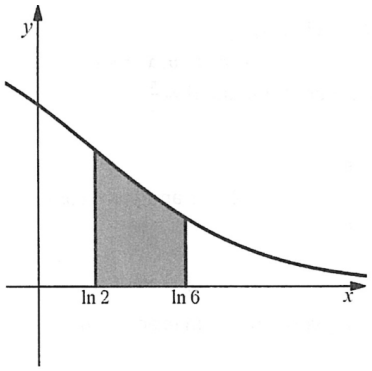
$$\int_0^{\pi} -45\sin^2\theta\, d\theta$$

c) Hence show that the shaded region has area $\frac{45\pi}{2}$.

2.43 — Question 18. The diagram shows the curve defined by the parametric equations

$$x = \ln(t-2) \qquad \text{and} \qquad y = \frac{6}{t}$$

for $t \geq 2$.



The shaded region is closed by the curve, the x-axis and the lines $x = \ln 2$ and $x = \ln 6$.

a) Show that the shaded area may be given by

$$\int_{t_1}^{t_2} \frac{6}{t(t-2)}\, dt$$

where t_1 and t_2 are constants to be found.

b) Show that

$$\frac{6}{t(t-2)} \equiv \frac{A}{t-2} + \frac{B}{t}$$

for some constants A and B.

c) Hence find the exact value of the shaded area in the form $\ln(\frac{a}{b})$ where a and b are positive integers.

2.43 — Question 19. The parametric equations of a curve are

$$x = \frac{2t^2}{2+t} \qquad \text{and} \qquad y = \frac{1}{2(2+t)}$$

where $t \geq 0$.

a) Find the coordinates of the curve when $x = 0$ and when $x = \frac{18}{5}$.

b) Show that the area enclosed by the curve, the x-axis and the line $x = 9$ is given by

$$\int_0^3 \frac{t(4+t)}{(2+t)^3}\,dt$$

c) Hence, by using the substitution $u = 2 + t$ find the exact area.

2.43 — Question 20. A curve is defined by parametric equations

$$x = t^2 \quad \text{and} \quad y = 1 - e^{-t}$$

for $t \geq 0$.

a) Find the y-value of the curve when $x = 0$ and when $x = 4$.

b) Sketch the curve for $0 \leq x \leq 4$.

c) Show that the area enclosed by the curve, the x-axis and the line $x = 4$ is given by

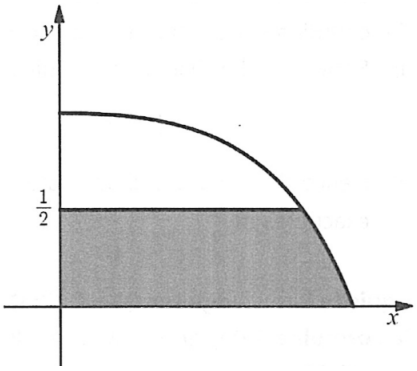
$$\int_0^2 (2t - 2te^{-t})\,dt$$

d) Hence, show the exact area is $2 + \frac{6}{e^2}$.

2.43 — Question 21. The diagram shows part of the curve defined by parametric equations

$$x = \frac{t}{t+1} \quad \text{and} \quad y = 1 - t^2$$

for $0 \leq t \leq 1$.



The shaded region is bounded by the curve, the coordinate axes and the line $y = \frac{1}{2}$.

a) Show that the shaded area is given by

$$\int_{\frac{\sqrt{2}}{2}}^{1} \frac{2t^2}{t+1}\, dt$$

b) Hence, by using the substitution $u = t + 1$ show that the shaded area has an exact area of

$$-\frac{3}{2} + \sqrt{2} + 2\ln(4 - 2\sqrt{2})$$

2.43 — Question 22. A curve is defined by the parametric equations

$$x = t + t^2 \qquad \text{and} \qquad y = k \sin t$$

where $0 \leq t \leq \pi$ and k is a positive constant. It is known that the area between the curve, the x-axis and the lines $x = 0$ and $x = 12$ is 10. Show that

$$k = \frac{10}{1 + 2\sin 3 - 7\cos 3}$$